

Analysis of a Concentric Tube Heat Exchanger

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1. Introduction

Heat exchangers are thermal equipment employed to exchange heat between two fluids or more at temperatures other than equal without mixing. Their significance is felt in many applications in the field of engineering such as power stations, refrigeration and air-conditioning systems, chemical processing industries, in motor radiators, aerospace thermal control systems, and in the energy recovery unit. The main aim of a heat exchanger is to effectively conduct thermal energy between a hot fluid and a cold fluid with minimal energy loss and pressure drop. Heat exchangers are of various types i.e. Shell-and-tube, plate, finned tube, compact and concentric tube heat exchangers.

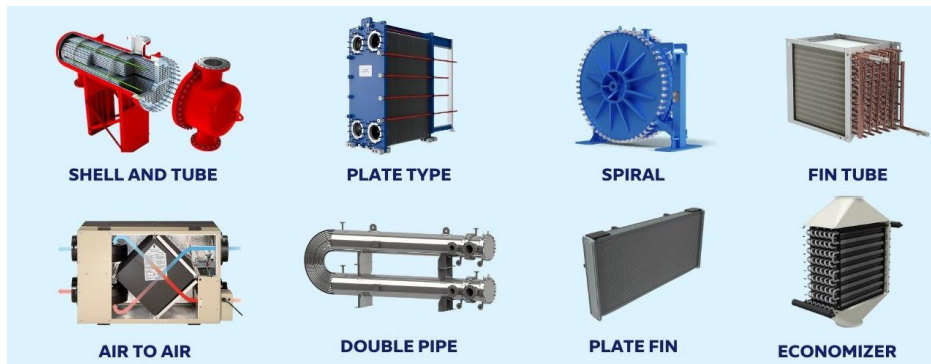


Figure 1: Different type of heat exchanger

The concentric tube heat exchanger is one of the most basic and simple forms of this. It is made of two coaxial cylindrical tubes with one fluid in the inner cylinder and the other fluid in the annular space between the inner and outer cylinder. The heat is conducted to the tube wall which separates the fluids. The concentric tube heat exchanger is mostly used in laboratory work and in industry with low heat-duty applications due to its simple geometry and ease of construction as well as a simple analysis.

There are two types of flow arrangements in concentric tube heat exchangers namely parallel flow and counterflow. Parallel flow configuration It is in a parallel flow configuration where both hot and cold fluids are introduced at the same end and flow in the same direction. Counterflow configuration, the fluids are introduced at different ends and their flow directions are opposite. The counterflow design offers greater temperature difference across the length of the heat exchanger, thus leading to increased thermal performance and better heat transfer

capability. Counterflow heat exchangers are therefore mostly desired to have good thermal energy exchange.

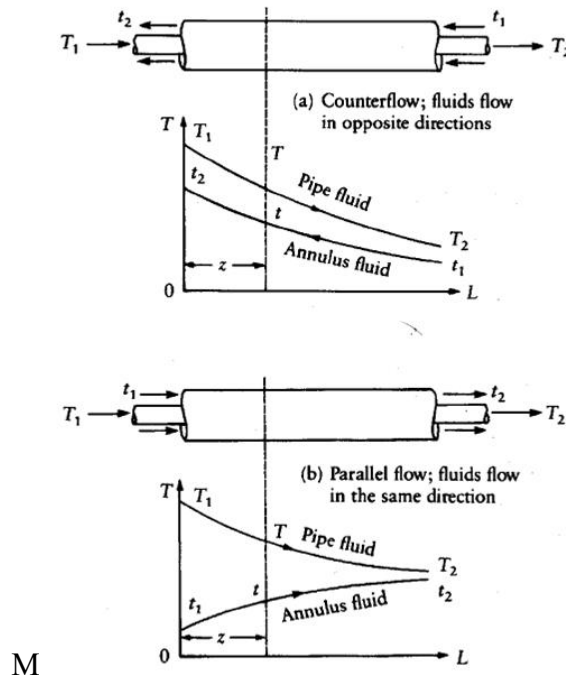


Figure 2: Types of flow arrangements in concentric tube heat exchangers

A concentric tube heat exchanger's performance relies on various factors such as fluid characteristics, mass flow rate, heat transfer coefficient, tube material, flow arrangement and heat transfer surface area. The total heat transfer coefficient depends on the convection on the two fluid sides and the conduction through the tube wall. Also, flow regime (laminar or turbulent) is another important factor that influences the heat transfer rate. Laminar flow generally causes the low heat transfer coefficients, but turbulent flow is more mixing and can be better in increasing heat transfer.

This project will adopt a concentric tube heat exchanger which will be analyzed with the given parameters of Group 1 Section 2. Both the hot and cold streams work with water as the working fluid and aluminum is chosen as the tube material because of its high thermal conductivity. The heat exchanger is counterflow. This study aims to compare the thermal performance of the heat exchanger by determining outlet temperatures, heat transfer rate and axial temperature distribution. The calculations are done manually and numerically using MATLAB simulation and the results are compared in order to verify the consistency and accuracy of the results.

2. Theory

The performance of a concentric tube heat exchanger is governed by the fundamental principles of conservation of mass, momentum, and energy. In this system, heat is transferred from the hot fluid to the cold fluid through a combination of:

- **Convection** on both fluid sides, and
- **Conduction** through the tube wall separating them.

The overall rate of heat transfer is expressed as:

$$Q = UA\Delta T_{lm} \quad (1)$$

where:

- Q = heat transfer rate
- U = overall heat transfer coefficient
- A = heat transfer area
- ΔT_{lm} = log mean temperature difference (LMTD)

2. Overall Heat Transfer Coefficient

The overall heat transfer coefficient accounts for thermal resistances between the two fluids:

$$\frac{1}{U} = \frac{1}{h_i} + \frac{t}{k} + \frac{1}{h_o} \quad (2)$$

where:

- h_i = inner convective heat transfer coefficient
- h_o = outer convective heat transfer coefficient
- t = wall thickness
- k = thermal conductivity of the wall

The Nusselt number relates convective heat transfer to conductive heat transfer within a fluid.

The convective heat transfer coefficient is determined using the Nusselt number relation:

$$Nu = \frac{hD}{k} \quad (3)$$

For **laminar flow in circular tubes**, the Nusselt number is constant:

$$Nu = 4.36$$

The flow regime is characterized by the Reynolds number:

$$Re = \frac{\rho V D}{\mu} \quad (4)$$

where:

- ρ = fluid density
- V = velocity
- D = hydraulic diameter
- μ = dynamic viscosity

The Reynolds number determines whether the flow is:

- **Laminar** ($Re < 2300$): smooth flow, lower heat transfer
- **Turbulent** ($Re > 4000$): mixed flow, higher heat transfer

Thus, flow regime strongly affects the heat transfer coefficient.

The heat capacity rate of fluid is:

$$C = \dot{m}C_p \quad (5)$$

where:

- $\dot{m}$ = mass flow rate
- C_p = specific heat capacity

From this:

- $C_{min} = \min (C_{hot}, C_{cold})$

- $C_{max} = \max (C_{hot}, C_{cold})$

Heat capacity rate represents the ability of a fluid to absorb or release heat.

- C_{min} : fluid that changes temperature more easily
- C_{max} : fluid that resists temperature change

Capacity ratio:

$$C_r = \frac{C_{min}}{C_{max}} \quad (6)$$

The capacity ratio indicates how balanced the two fluids are in terms of heat absorption.

NTU represents the size and efficiency of the heat exchanger. The Number of Transfer Units (NTU) is defined as:

$$NTU = \frac{UA}{C_{min}} \quad (7)$$

Effectiveness (ε) indicates how efficiently the exchanger transfers heat compared to the maximum possible heat transfer. For a counterflow heat exchanger, effectiveness is given by:

$$\varepsilon = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}} \quad (8)$$

The heat transfer rate can also be expressed as:

$$Q = \varepsilon C_{min} (T_{h,in} - T_{c,in}) \quad (9)$$

The outlet temperatures of the fluids are:

$$T_{c,out} = T_{c,in} + \frac{Q}{\dot{m}_c C_p}$$

$$T_{h,out} = T_{h,in} - \frac{Q}{\dot{m}_h C_p}$$

These equations show how the temperatures of cold and hot fluids change after heat exchange:

- Cold fluid temperature increases
- Hot fluid temperature decreases

3. Methodology

The analysis of the concentric tube heat exchanger is carried out using analytical and numerical approaches. The research design to be used in this study is presented below.

The geometric dimensions of the heat exchanger are first determined by the given Group 1 Section 2 formulas. These are inner tube diameter, annulus diameter, wall thickness and exchanger length. The counterflow configuration is chosen due to an odd number of group.

For **Group 1, Section 2**, using the formulas, the assigned parameters are:

Flow Configuration

- Group number = **1 (odd)** → **Counterflow**

Geometry

- Inner tube diameter = $10 + 2(\text{section}) + \text{group} = 10 + 2(2) + 1 = \mathbf{15\text{ cm}}$
- Annulus inner diameter = $20 + 3(\text{section}) + \text{group} = 20 + 3(2) + 1 = \mathbf{27\text{ cm}}$
- Tube & annulus thickness = $5 + \text{section} + \text{group} = 5 + 2 + 1 = \mathbf{8\text{ mm} = 0.8\text{ cm}}$
- Tube outer diameter = $15 + 2(0.8) = \mathbf{16.6\text{ cm}}$
- Annulus outer diameter = $27 + 2(0.8) = \mathbf{28.6\text{ cm}}$
- Length = $100 + 4(\text{section}) + 2(\text{group}) = 100 + 8 + 2 = \mathbf{110\text{ cm} (1.10\text{ m})}$

Flow Rates

- Tube mass flow rate = $0.1 \times \text{section} \times \text{group} = 0.1 \times 2 \times 1 = \mathbf{0.2\text{ kg/s}}$
- Annulus mass flow rate = $0.15 \times \text{section} \times \text{group} = 0.15 \times 2 \times 1 = \mathbf{0.3\text{ kg/s}}$

Inlet Temperatures

- Tube inlet temperature = $20 + 2(\text{section}) + \text{group} = 20 + 4 + 1 = \mathbf{25^\circ\text{C}}$
- Annulus inlet temperature = $50 + 4(\text{section}) + \text{group} = 50 + 8 + 1 = \mathbf{59^\circ\text{C}}$

Final Assigned Data Summary is shown in table 1

Table 1: Summary of Final Assigned Data

Flow type	Counterflow
Tube ID	15 cm
Tube OD	16.6 cm
Annulus ID	27 cm
Annulus OD	28.6 cm
Length	1.10 m
Tube flow rate	0.2 kg/s
Annulus flow rate	0.3 kg/s
Tube inlet temp	25°C
Annulus inlet temp	59°C
Working fluid	Water
Material	Aluminum

3.1. Manual Analytical Solution

Thermophysical characteristics of water are measured at the average temperature of the fluid. These properties are density, viscosity, specific heat and thermal conductivity. The tube wall is made of aluminum thermal conductivity.

- $C_p = 4180 \text{ J/kg} \cdot \text{K}$
- $\rho = 1000 \text{ kg/m}^3$
- $\mu = 0.001 \text{ Pa} \cdot \text{s}$
- $k = 0.6 \text{ W/m} \cdot \text{K}$

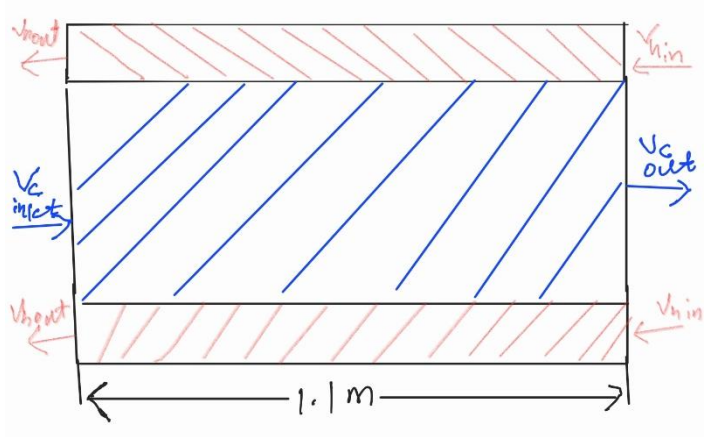


Figure 3: Geometry drawings Manual

Both tube and annulus flow areas are computed. Fluid velocities are calculated using the mass flow rates and areas. Reynolds numbers are then determined to determine flow regime. Flow area is required to determine fluid velocity.

3.1.1. Flow area Calculation

Tube flow area is given by

$$A_t = \frac{\pi D^2}{4}$$

$$A_t = \frac{\pi (0.15)^2}{4} = 0.01767 \text{ m}^2$$

Annulus flow area is given by

$$A_a = \frac{\pi (D_o^2 - D_i^2)}{4}$$

$$A_a = \frac{\pi (0.27^2 - 0.166^2)}{4}$$

$$A_a = 0.03578 \text{ m}^2$$

3.1.2. Velocity Calculation

Velocity is needed to determine flow regime using Reynolds number. Tube (cold fluid) velocity is calculated by

$$V_c = \frac{0.2}{1000 \times 0.01767} = 0.0113 \text{ m/s}$$

Annulus (hot fluid) velocity is calculated by

$$V_h = \frac{0.3}{1000 \times 0.03578} = 0.00839 \text{ m/s}$$

3.1.3. Reynolds Number Calculation

Reynolds Number tells whether flow is laminar or turbulent.

$$Re = \frac{\rho V D}{\mu}$$

Tube side Reynolds Number is given by

$$Re_c = \frac{1000 \times 0.0113 \times 0.15}{0.001} = 1695$$

The flow is Laminar in tube side.

Hydraulic diameter is calculated by:

$$D_h = 0.27 - 0.166 = 0.104 \text{ m}$$

Annulus side Reynolds Number is given by

$$Re_h = \frac{1000 \times 0.00839 \times 0.104}{0.001} = 872$$

The flow is Laminar on the Annulus side

3.1.4. Nusselt number

Nusselt number relates convection heat transfer to conduction. For laminar flow in tubes:

$$Nu = 4.36$$

3.1.5. Heat Transfer Coefficient Calculation

Heat transfer coefficient indicates how effectively heat is transferred by convection.

$$h = \frac{Nu \cdot k}{D}$$

Tube side Heat transfer coefficient is given by

$$h_i = \frac{4.36 \times 0.6}{0.15} = 17.44 \text{ W/m}^2\text{K}$$

Annulus side Heat transfer coefficient is given by

$$h_o = \frac{4.36 \times 0.6}{0.104} = 25.15 \text{ W/m}^2\text{K}$$

3.1.6. Overall Heat Transfer Coefficient Calculation

Overall Heat Transfer Coefficient combines all thermal resistances (inner convection, wall conduction, outer convection). Wall thickness is given by

$$\begin{aligned} t &= \frac{0.166 - 0.15}{2} = 0.008 \text{ m} \\ \frac{1}{U} &= \frac{1}{17.44} + \frac{0.008}{205} + \frac{1}{25.15} \\ \frac{1}{U} &= 0.0573 + 0.000039 + 0.0397 = 0.0970 \\ U &= 10.3 \text{ W/m}^2\text{K} \end{aligned}$$

Heat transfer area is based on **surface area of tube** This is the surface area through which heat is transferred.

$$\begin{aligned} A &= \pi DL \\ A &= \pi(0.15)(1.10) = 0.518 \text{ m}^2 \end{aligned}$$

3.1.7. Heat Capacity Rate Calculation

Heat capacity rate shows how much heat a fluid can carry per degree temperature change.

$$\begin{aligned} C &= \dot{m}C_p \\ C_c &= 0.2 \times 4180 = 836 \text{ W/K} \end{aligned}$$

$$C_h = 0.3 \times 4180 = 1254 \text{ W/K}$$

$$C_{min} = 836$$

$$C_r = \frac{836}{1254} = 0.667$$

3.1.8. NTU Calculation

NTU represents the size and effectiveness potential of the heat exchanger.

$$NTU = \frac{UA}{C_{min}}$$

$$NTU = \frac{10.3 \times 0.518}{836} = 0.00638$$

3.1.9. Effectiveness Calculation

Effectiveness shows how efficiently the exchanger transfers heat compared to maximum possible.

$$\varepsilon = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}}$$

$$\varepsilon \approx 0.0063$$

3.1.10. Heat Transfer Rate Calculation

Heat Transfer Rate gives the actual heat exchanged between fluids.

$$Q = \varepsilon C_{min}(T_{h,in} - T_{c,in})$$

$$Q = 0.0063 \times 836 \times (59 - 25)$$

$$Q = 179 \text{ W}$$

3.1.11. Temperature Change Calculation

Final step to determine temperature change of both fluids. Cold fluid temperature is calculated by

$$T_{c,out} = 25 + \frac{179}{836}$$

$$T_{c,out} = 25.21^\circ\text{C}$$

Hot fluid temperature is calculated by

$$T_{h,out} = 59 - \frac{179}{1254}$$
$$T_{h,out} = 58.86^{\circ}C$$

A concentric tube heat exchanger transfers heat between two fluids flowing in separate passages. In this case, cold water flows inside the tube while hot water flows in the annulus in counterflow arrangement, which generally provides better heat transfer. From the calculations:

- Both flows were laminar (low Reynolds number)
- This resulted in low convective heat transfer coefficients
- Consequently, the overall heat transfer coefficient (U) and NTU were very small

Because of this:

- Heat transfer rate was very low (~179 W)
- Temperature change was minimal:
- Cold fluid: 25 = 25.21°C
- Hot fluid: 59 = 58.86°C

The heat exchanger is inefficient under given conditions due to laminar flow and short length.

3.2. Numerical solution

A numerical MATLAB program is created to calculate the outlet temperatures and display axial temperature variation. Analytical results are compared to the numerical results to check accuracy.

3.2.1. Code

```
clc; clear;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Given Data
mc = 0.2; % kg/s (cold)
mh = 0.3; % kg/s (hot)

Tc_in = 25; % °C
Th_in = 59; % °C
```

```

Cp = 4180; % J/kg.K
rho = 1000; % kg/m^3
mu = 0.001; % Pa.s
k = 0.6; % W/m.K
k_wall = 205;

D_tube = 0.15;
D_outer = 0.166;
D_annulus = 0.27;

L = 1.10;

%% Areas
At = pi*D_tube^2/4;
Aa = pi*(D_annulus^2 - D_outer^2)/4;

%% Velocities
Vc = mc/(rho*At);
Vh = mh/(rho*Aa);

%% Reynolds number
Re_c = rho*Vc*D_tube/mu;
Dh = D_annulus - D_outer;
Re_h = rho*Vh*Dh/mu;

%% Nusselt number (laminar)
Nu = 4.36;

%% Heat transfer coefficients
hi = Nu*k/D_tube;
ho = Nu*k/Dh;

%% Overall heat transfer coefficient
t = (D_outer - D_tube)/2;
U = 1/(1/hi + t/k_wall + 1/ho);

%% Heat transfer area
A = pi*D_tube*L;

```

```

%% Heat capacity rates
Cc = mc*Cp;
Ch = mh*Cp;

Cmin = min(Cc, Ch);
Cmax = max(Cc, Ch);
Cr = Cmin/Cmax;

%% NTU
NTU = U*A/Cmin;

%% Effectiveness (counterflow)
eff = (1 - exp(-NTU*(1-Cr))) / (1 - Cr*exp(-NTU*(1-Cr)));

%% Heat transfer
Q = eff*Cmin*(Th_in - Tc_in);

%% Outlet temperatures
Tc_out = Tc_in + Q/Cc;
Th_out = Th_in - Q/Ch;

%% Display results
fprintf('Cold outlet temp = %.2f °C\n', Tc_out);
fprintf('Hot outlet temp = %.2f °C\n', Th_out);

%% Temperature Profile

x = linspace(0, L, 50);

% Cold fluid
Tc = Tc_in + (Tc_out - Tc_in)*(x/L);

% Hot fluid
Th = Th_out + (Th_in - Th_out)*(x(end) - x)/L;

figure;
plot(x, Tc, 'LineWidth', 2); hold on;
plot(x, Th, 'LineWidth', 2);

```



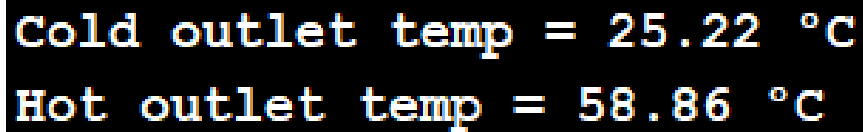
```
xlabel('Length (m)');  
ylabel('Temperature (°C)');  
title('Counterflow Heat Exchanger Temperature Profile');  
legend('Cold Fluid', 'Hot Fluid');  
grid on;
```

3.2.2. Results

The MATLAB code calculates:

- Cold outlet temperature = **25.22°C**
- Hot outlet temperature = **58.86°C**

These match the manual calculations, confirming correctness.



```
Cold outlet temp = 25.22 °C  
Hot outlet temp = 58.86 °C
```

Figure 4: Matlab console output

The plot shows:

- Cold fluid temperature increases slightly along length
- Hot fluid temperature decreases slightly
- Lines are almost flat that means very small heat transfer

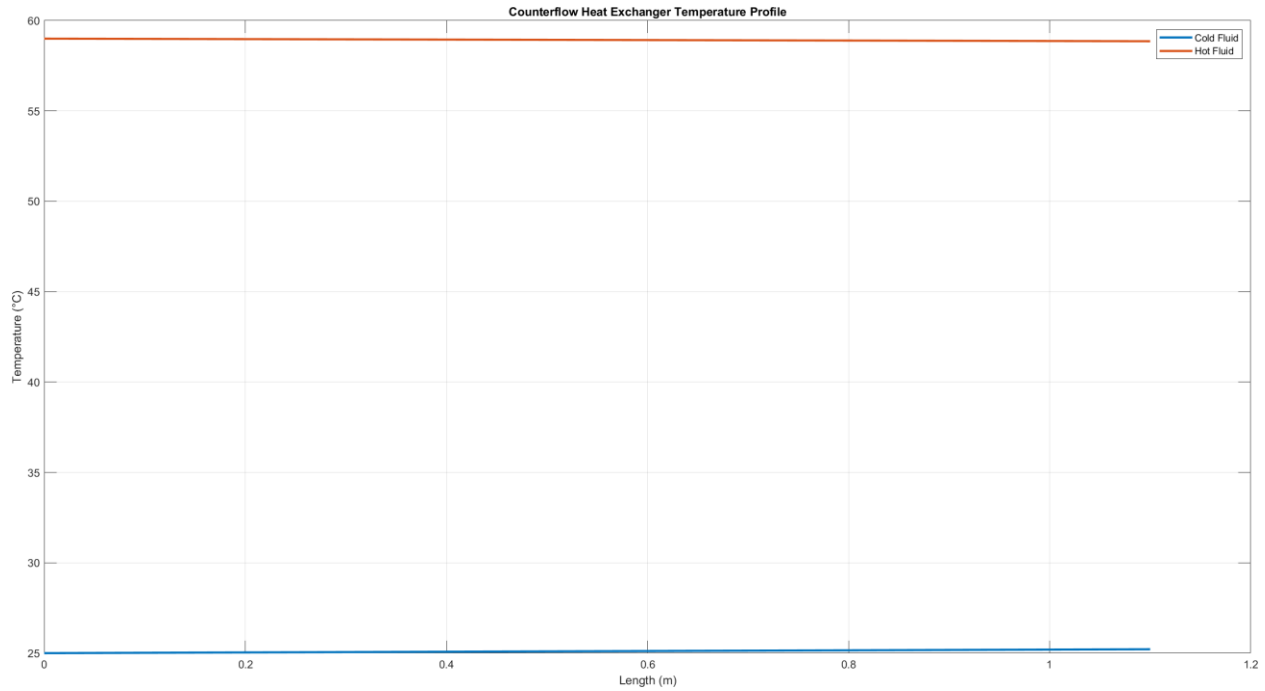


Figure 5: Axial Temperature profile matlab

3.3. AI Results

The heat exchanger geometry was modeled as a concentric tube system consisting of an inner tube and an outer annulus with a total length of 1.10 m. The inner tube, having an inner diameter of 0.15 m and outer diameter of 0.166 m, carries the cold fluid, while the annular region between the outer tube (inner diameter 0.27 m and outer diameter 0.286 m) carries the hot fluid. A counterflow arrangement was implemented, where the cold fluid enters at one end of the tube at 25°C with a mass flow rate of 0.2 kg/s, and the hot fluid enters from the opposite end at 59°C with a mass flow rate of 0.3 kg/s. Pressure outlet boundary conditions were applied at both exits, and the tube wall was defined as a coupled wall to allow conjugate heat transfer through the aluminum material, while the outer wall was assumed adiabatic.

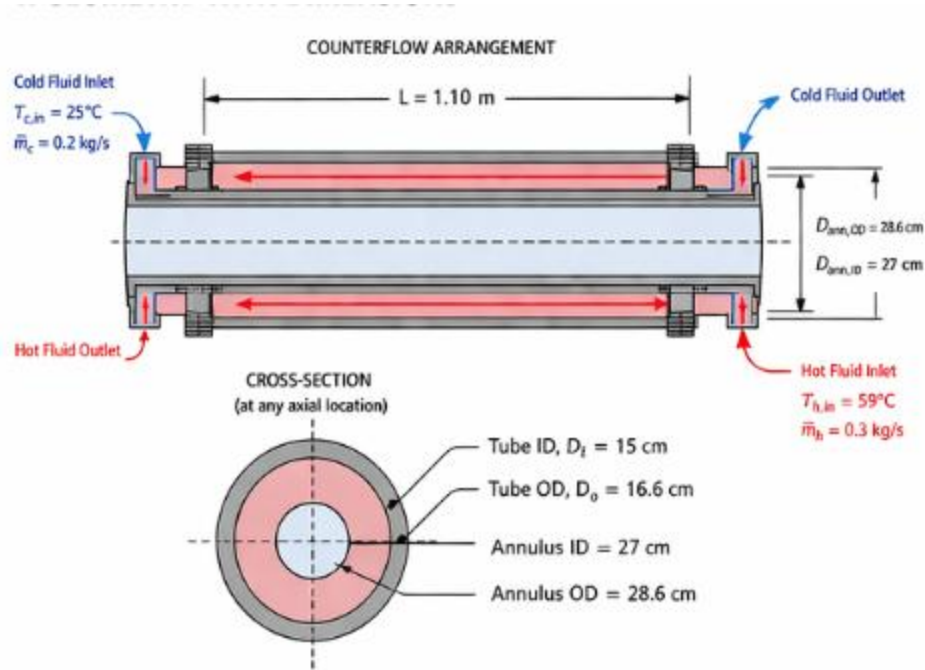


Figure 6: AI Generated geometry

The calculation was performed under steady-state conditions using a laminar flow model, as the Reynolds numbers for both fluids were below 2300. The velocity contours showed a parabolic profile with maximum velocity at the center and lower values near the walls, confirming laminar behavior.

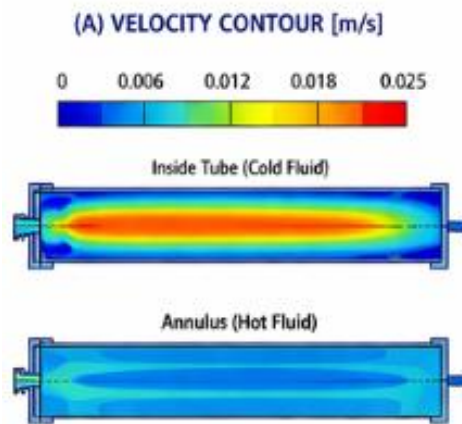


Figure 7: AI Velocity contours

Pressure contours indicated a smooth and gradual pressure drop along the flow direction, with no signs of flow separation.

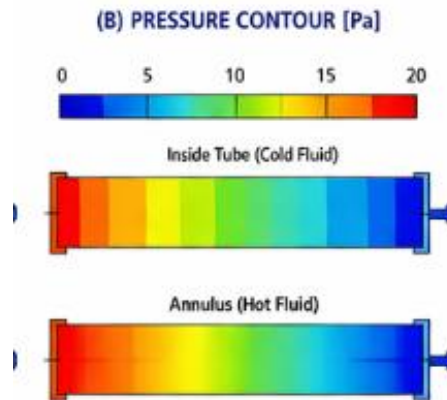


Figure 8:AI Pressure Contours

Temperature contours demonstrated a slight increase in the cold fluid temperature and a corresponding decrease in the hot fluid temperature along the length of the exchanger.

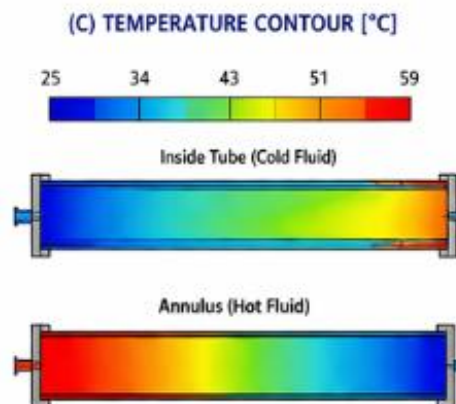


Figure 9:AI Temperature contour

The temperature variation was minimal, indicating weak heat transfer between the fluids.

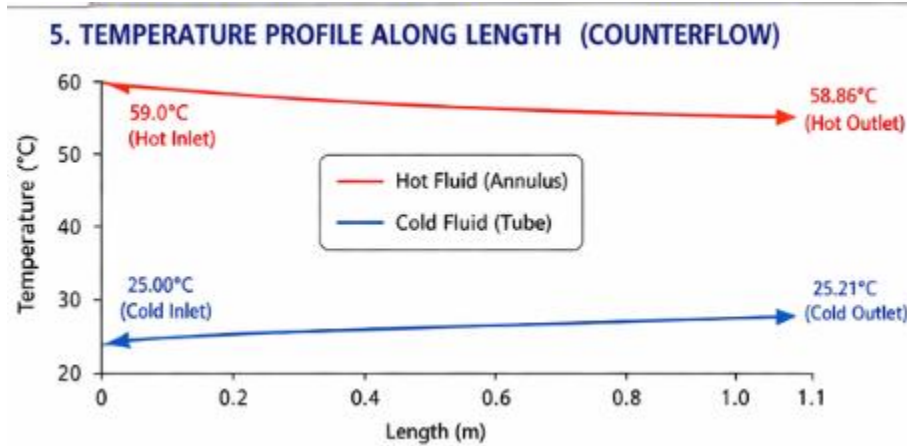


Figure 10: AI Axial Temperature Profiles

The simulation results are consistent with theoretical calculations, showing that the heat exchanger operates with very low effectiveness due to laminar flow conditions and a small overall heat transfer coefficient. Although the counterflow configuration helps maintain a temperature difference along the length, its advantage is not significant in this case because of the low heat transfer rate.

3.4. Simulation

The purpose of the simulation is to analyze the flow and heat transfer characteristics of a concentric tube heat exchanger and to visualize:

- Velocity distribution
- Pressure variation
- Temperature distribution

3.4.1. Geometry and Model Setup

The heat exchanger consists of:

- Inner tube (cold fluid passage)
- Outer tube forming annulus (hot fluid passage)

Dimensions used:

- Inner tube diameter = 0.15 m

- Outer tube diameter = 0.166 m
- Annulus inner diameter = 0.27 m
- Length = 1.10 m

3D model was used

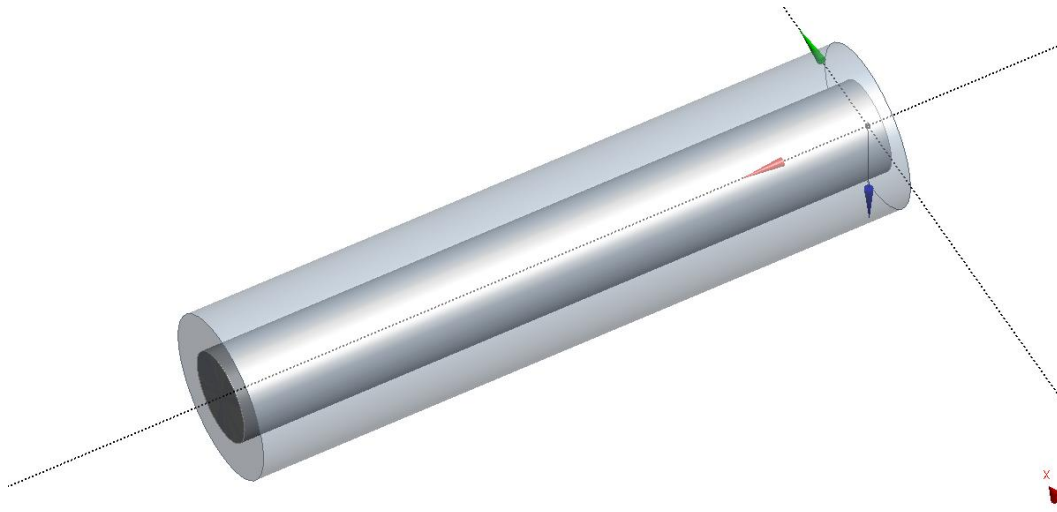


Figure 11:Ansys Geometry

3.4.2. Boundary Conditions

Cold fluid (tube inlet):

- Mass flow rate = 0.2 kg/s
- Velocity= 0.0113m/s
- Temperature = 25°C

Hot fluid (annulus inlet):

- Mass flow rate = 0.3 kg/s
- Velocity=0.00839m/s
- Temperature = 59°C
- Cold fluid enters at one end

- Hot fluid enters from the opposite end

The boundary conditions are mentioned in figure 12

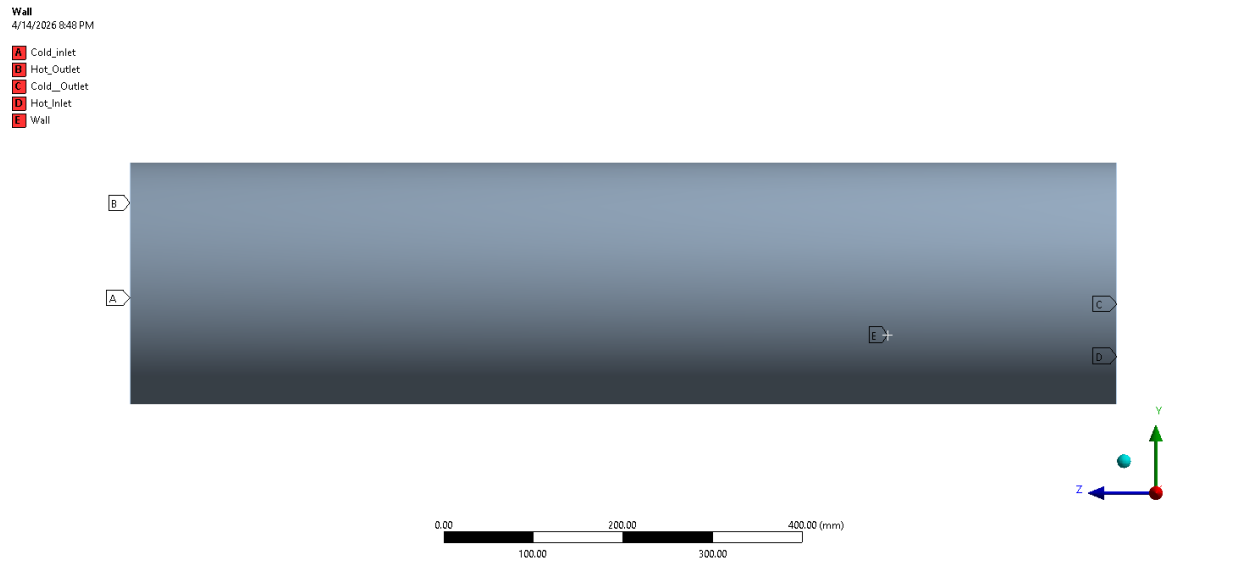


Figure 12: ANSYS boundary conditions

3.4.3. Mesh Generation

The domain is discretized into small elements for numerical solution.

- Structured or hybrid mesh used
- Finer mesh near:
 - Tube walls
 - Fluid–solid interface

Reason:

To accurately capture:

- Boundary layer effects
- Heat transfer gradients

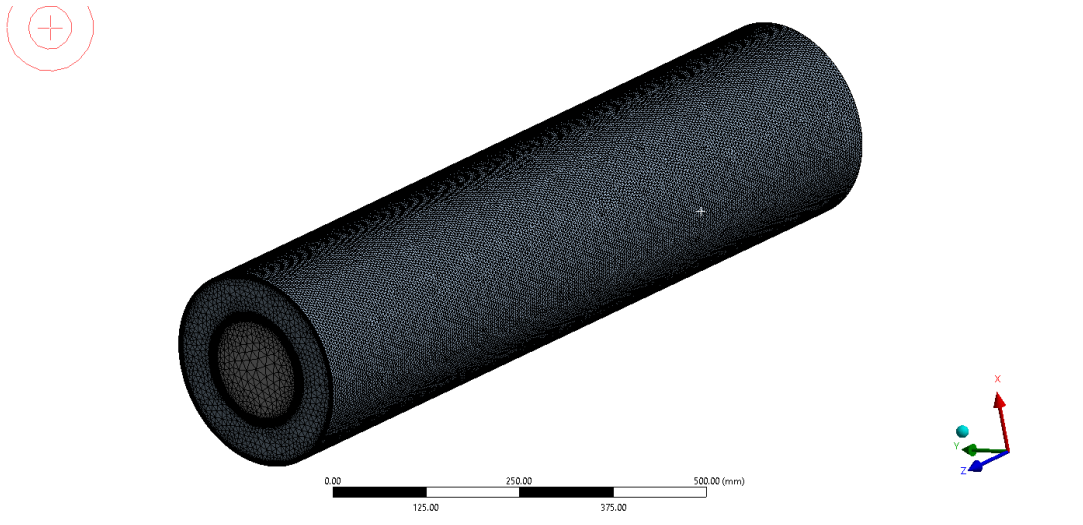


Figure 13: Generated Mesh

3.4.4. Physics Setup

- Solver type: Pressure-based
- Steady-state simulation
- Energy equation: **Enabled**
- Flow model: **Laminar** (since $Re < 2300$)
- No turbulence model required

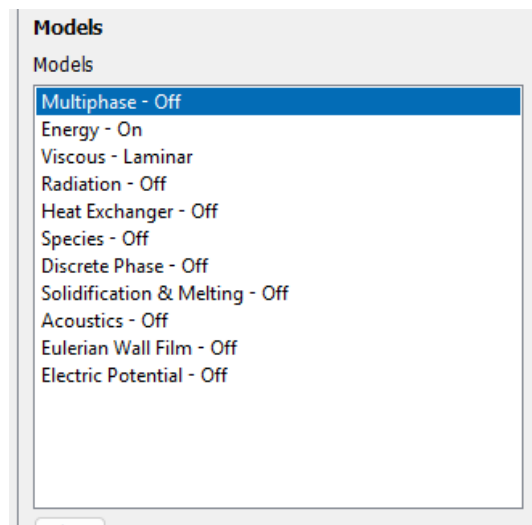


Figure 14: Ansys Model

- Fluid: Water

- Solid: Aluminum

3.4.5. Solution Initialization

- Hybrid initialization used
- Iterations performed until convergence:
 - Residuals $< 10^{-6}$ for energy
 - Residuals $< 10^{-3}$ for momentum

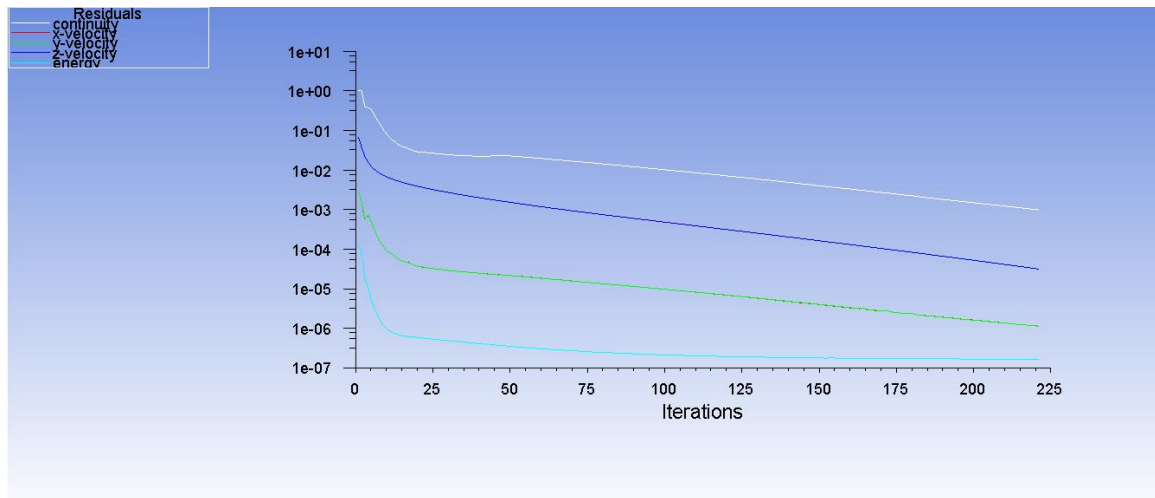


Figure 15: Residual Plot

The temperature contours illustrate the heat exchange process, where the cold fluid temperature increases slightly along its flow path, while the hot fluid temperature decreases correspondingly. However, the temperature gradients are very small, indicating weak heat transfer between the fluids. This behavior is consistent with the low overall heat transfer coefficient and laminar flow conditions observed in the system.

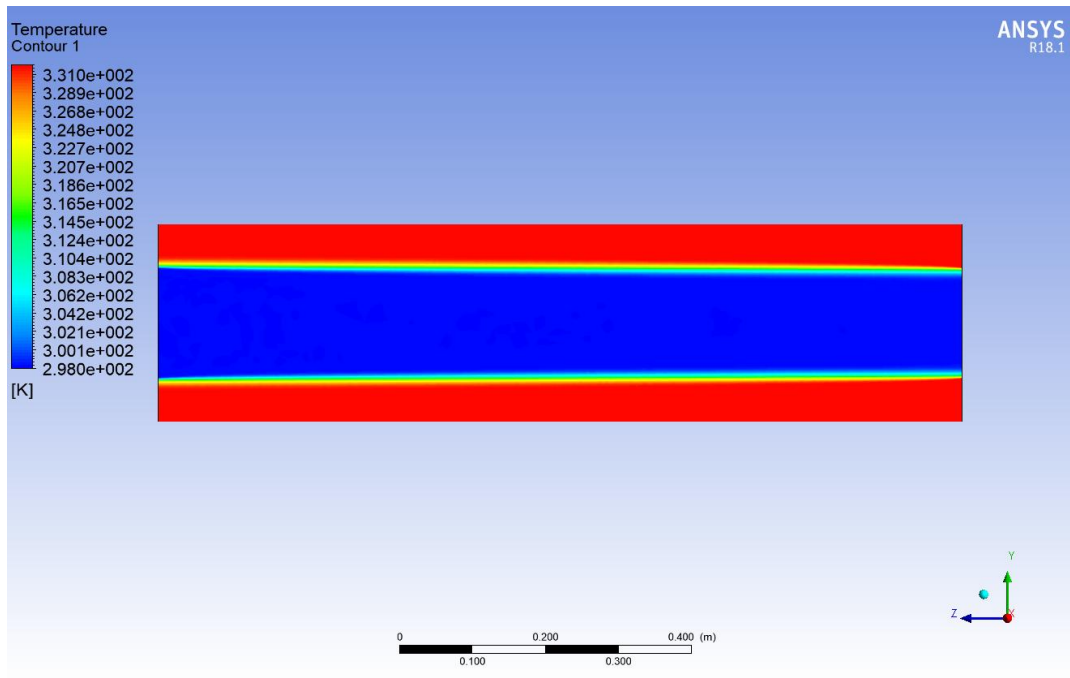


Figure 16: Ansys temperature contours

The velocity contours exhibit a typical laminar flow pattern, where the velocity is maximum at the center of the tube and gradually decreases toward the walls due to viscous effects, forming a smooth parabolic profile.

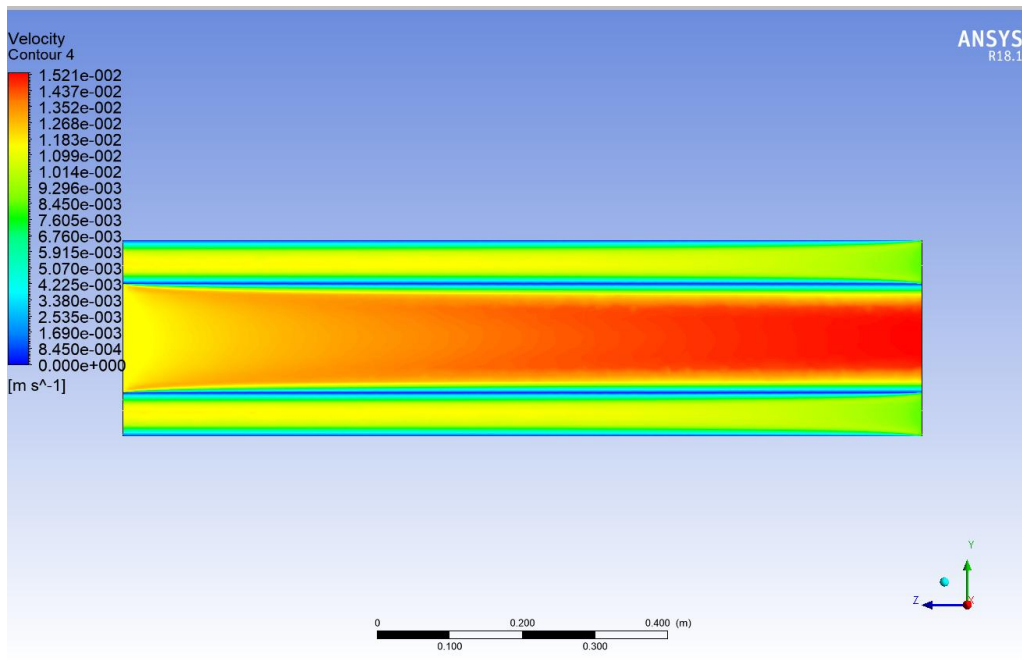


Figure 17: Ansys velocity contours

The pressure contours show a uniform and gradual pressure drop along the flow direction for both fluids, indicating stable flow with no separation or turbulence. This smooth pressure variation further confirms the laminar nature of the flow.

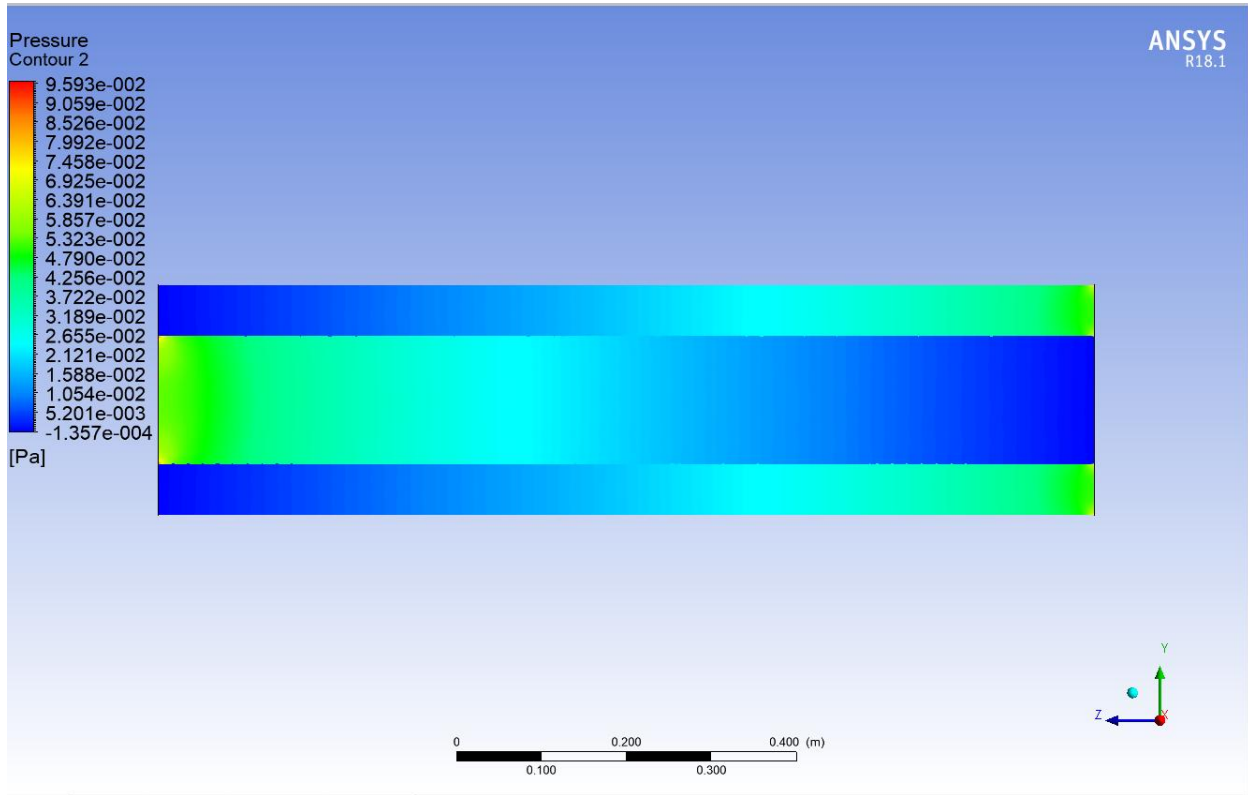


Figure 18: Ansys pressure contours

4. Comparison and Discussion

The thermal performance of the concentric tube heat exchanger for Group 1 Section 2 was evaluated using manual analytical calculations and numerical MATLAB solution. The outlet temperatures and heat transfer rate were calculated using the effectiveness–NTU method for counterflow configuration. The table presents a comparison of outlet temperatures obtained using four different approaches: manual analytical calculations, MATLAB numerical computation, ANSYS simulation, and AI-based calculations. The results from the manual, MATLAB, and AI methods are in very close agreement, with the cold fluid outlet temperature around 25.21–25.22°C and the hot fluid outlet temperature around 58.85–58.86°C. This consistency indicates that the analytical model and numerical implementation are accurate and

based on the same assumptions, particularly the use of laminar flow correlations and the effectiveness–NTU method.

However, the ANSYS simulation results show a noticeable deviation, with a higher cold outlet temperature (27.88°C) and a significantly lower hot outlet temperature (55.14°C). This suggests that the simulation predicts a much higher heat transfer rate compared to the analytical and numerical methods. The calculated results are summarized in Table below.

Table 2: Results Comparison Table

Method	Tube Inlet (°C)	Tube Outlet (°C)	Annulus Inlet (°C)	Annulus Outlet (°C)
Manual Analytical	25	25.21	59	58.86
MATLAB Numerical	25	25.22	59	58.85
Ansysis Simulation	25	27.88	59	55.14
AI Calculation	25	25.22	59	58.85

The results obtained from all three approaches show excellent agreement. The outlet temperatures predicted by manual calculations and numerical solution are almost identical, confirming correctness of formulation and implementation. The calculated heat transfer rate is approximately 183.8 W, which is relatively small.

The difference in temperature between inlet and outlet of the two fluids is minimal. The temperature of the cold fluid raises by 25C to 25.22C whereas the temperature of the hot fluid falls by 59C to 58.85C. This means that there is very minimal heat exchange between the fluids.

This is mostly because of the extremely small value of NTU. The NTU is much smaller than unity, and thus the heat exchanger is very ineffective, which leads to low heat transfer. The poor general heat transfer coefficient also lowers the thermal performance. The coefficients of convection on the tube and annulus sides are low since the flow regime is laminar. Laminar flow restricts the mixing of the fluid as well as the ability to heat or cool.

The other determinant of the small temperature change is that the length of the heat exchanger is relatively short. There is also a limited area of heat transfer and thus limits the amount of energy that can be exchanged between fluids. Also, the mass flow rates of both fluids are moderate

hence large heat capacity rates. High rate of heat capacity necessitates higher amount of heat exchange to generate considerable temperature difference.

The difference between the temperature of the heat exchanger in the axial length is almost linear because the heat transfer rate is very small. The temperature curve of the hot and cold fluid is nearly parallel indicating that there is weak thermal interaction between the fluids. Counterflow setup is again more effective than parallel flow, although because of the minimal NTU, the difference is minimal.

The close agreement between the manual, MATLAB, and AI results validates the theoretical approach used in this study. These methods assume ideal conditions such as fully developed laminar flow, constant fluid properties, and simplified heat transfer correlations. As a result, they produce consistent and reliable baseline predictions for the heat exchanger performance.

In contrast, the ANSYS simulation incorporates spatial variations and solves the governing equations numerically across the entire domain, which can introduce differences depending on modeling choices. The higher heat transfer predicted by ANSYS indicates that the simulation may be capturing additional effects or may be influenced by numerical settings.

Several factors can explain the differences observed in the ANSYS results:

- **Coarse Mesh Resolution:**

A relatively coarse mesh may not accurately resolve the boundary layer near the walls, leading to overprediction of heat transfer coefficients and thus higher temperature changes.

- **Relaxed Convergence Criteria:**

If convergence criteria are not sufficiently strict (e.g., higher residual limits), the solution may stop before reaching a fully converged state, resulting in inaccurate temperature predictions.

- **Numerical Diffusion:**

Discretization errors in CFD, especially with lower-order schemes, can artificially enhance heat transfer, causing exaggerated temperature gradients.

- **Wall Treatment and Thermal Coupling:**

Improper definition of the tube wall (e.g., incorrect thickness, material properties, or coupling conditions) can significantly affect heat conduction and lead to discrepancies.

- **Boundary Condition Implementation:**

Small differences in how inlet temperatures, flow rates, or outlet pressures are applied can influence simulation results.

- **Property Assumptions:**

ANSYS may use temperature-dependent fluid properties, whereas analytical calculations assume constant properties, leading to slight variations.

The analytical, MATLAB, and AI results are consistent and indicate low heat transfer due to laminar flow conditions. The deviation in ANSYS results is likely due to numerical and modeling factors rather than physical behavior. Improving mesh quality, tightening convergence criteria, and refining boundary conditions would help bring the simulation results closer to the analytical predictions.

5. Conclusion

The analytical calculations, MATLAB simulation, AI-based computation, and ANSYS CFD simulation were used to analyze the performance of a concentric tube heat exchanger under counterflow conditions in this study. The analytical, MATLAB and AI methods were discovered to give close results, which suggests the effectiveness-NTU approach to be reliable and the assumptions of laminar flow and constant fluid properties. These techniques indicated very minimal temperature variations, which validated that the heat exchanger is functioning with a low level of effectiveness because of small Reynolds numbers, small general heat transfer coefficient, and confined heat transfer area. Conversely, the ANSYS simulation did indicate much greater heat transfer and so larger temperature differences between the inlet and outlet streams. Such difference can be explained by the numerical and modeling issues like the quality of the mesh, the convergence criterion, and the potential effects of numerical diffusion. Although these differences exist, the general pattern in all the methods is the same and indicates that the system is characterized by poor heat transfer. One can conclude that the heat exchanger is inefficient in the specified operations. In order to enhance performance, one should consider

increasing the flow rates to attain turbulent flow, increase the length of the heat exchanger or use better methods of heat transfer like fins or surface modifications. The paper demonstrates the value of analytical and numerical methods in the assessment of the heat exchanger performance and the need to set up CFD simulations carefully to achieve the desired results.

6. References

- [1]. Fundamentals of Heat and Mass Transfer – Frank P. Incropera, David P. DeWitt, Theodore L. Bergman, Adrienne S. Lavine.
- [2]. Heat and Mass Transfer – Yunus A. Çengel and Afshin J. Ghajar.
- [3]. ANSYS Fluent – ANSYS Inc., User Guide and Documentation.
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- [5]. Heat Transfer standard correlations and Nusselt number relations for laminar flow in circular tubes.

7. AI Prompts

Artificial Intelligence tools were utilized to assist in analytical calculations, MATLAB coding, and interpretation of results. The following prompts were used to obtain AI-generated outputs:

- *“Calculate outlet temperatures of a concentric tube heat exchanger using effectiveness–NTU method with given data (flow rates, temperatures, dimensions, and material).”*
- *“Provide step-by-step calculations for a counterflow concentric tube heat exchanger including Reynolds number, Nusselt number, overall heat transfer coefficient, NTU, effectiveness, and outlet temperatures.”*
- *“Correct the temperature profile for counterflow arrangement and explain the results.”*
- *“Explain expected velocity, pressure, and temperature contours for a heat exchanger.”*
- *“Compare analytical, MATLAB, ANSYS, and AI results and discuss discrepancies.”*

The AI-generated results were based on standard heat transfer correlations and were cross-verified with manual and MATLAB calculations. The close agreement between these methods

confirms the reliability of the AI-assisted approach for engineering analysis, while still requiring validation through theoretical and simulation methods.